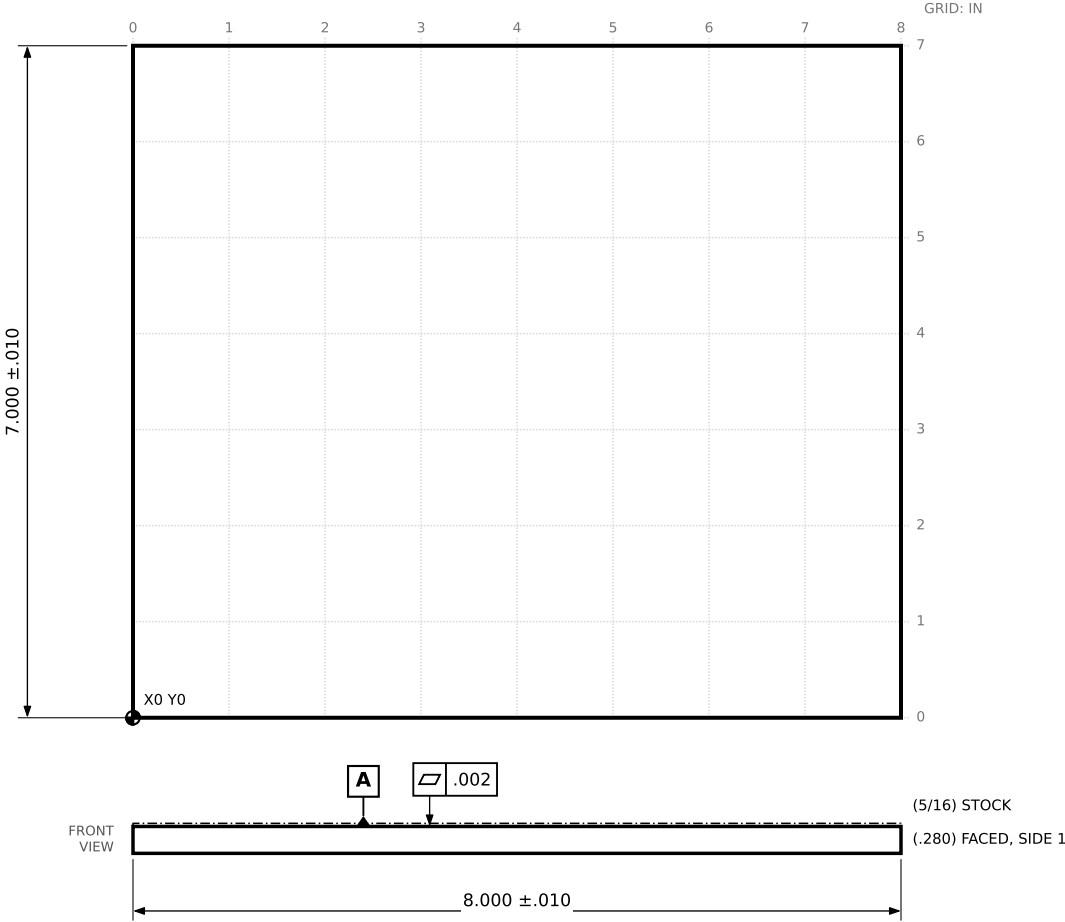


PEN PLOTTER: GANTRY END PLATE, OP1 FACE SIDE 1 (INNER FACE)

O60011



NO HOLES IN THIS SETUP: FACING ONLY

FEATURE CALLOUTS

FACE (T16)
FACE THE INNER FACE (DATUM A)
STOCK 5/16, FACE TO (.280) REF
OP2 FACES TO .250 ±.005
T16: 1 ROUGH PASS + .005 FINISH

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
.XX ±.01 .XXX ±.005 .XXXX ±.0005
ANGLES ±0.5°
BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 µin Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

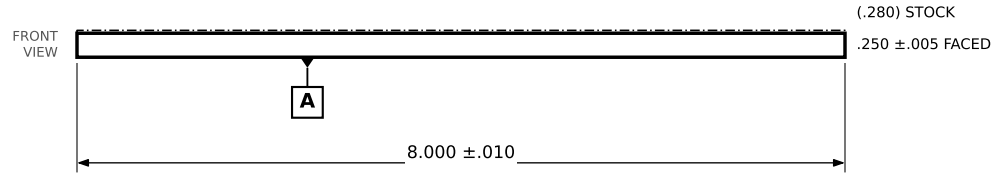
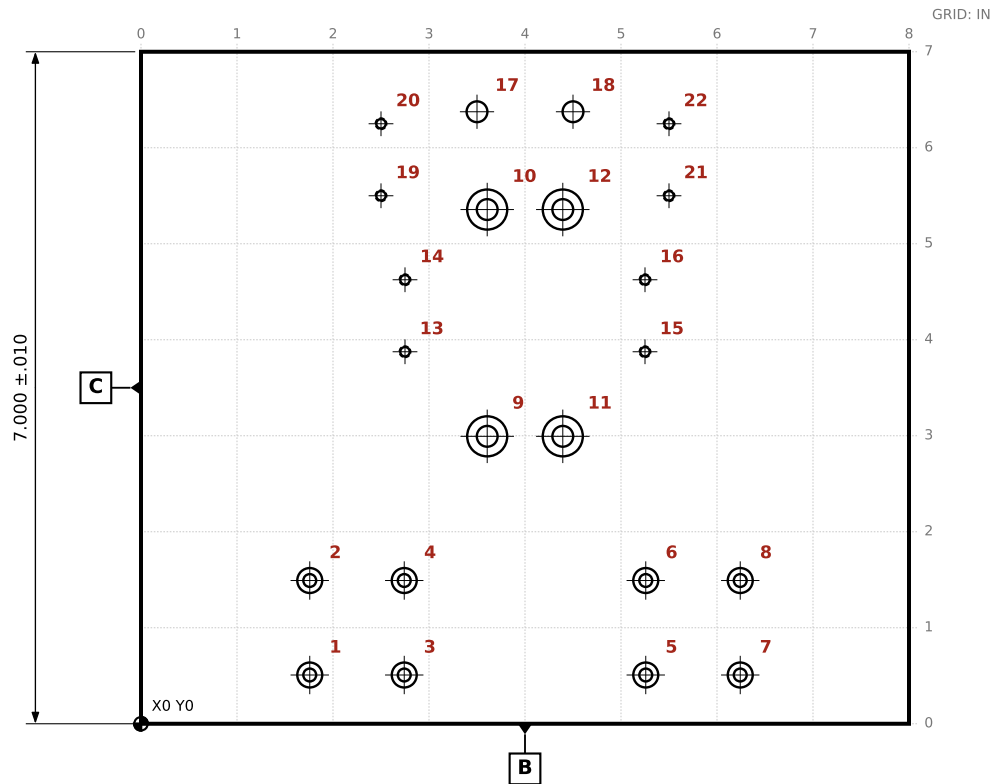
NOTES

1. FIXTURE: 5/16 PLATE BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE LONG EDGES, INNER FACE UP.
2. ORIENT: LONG SIDE 8.000 ALONG X.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .280 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS.
4. DATUMS: A = INNER FACE, AGAINST THE MGN15H BLOCKS (FACED IN OP1, UP); B = EDGE AT PROGRAM Y0 IN OP2; C = EDGE AT PROGRAM X0 IN OP2.

PART	GANTRY END PLATE			
PART NO.	END_PLATE			
SETUP	OP1 FACE SIDE 1 (INNER FACE)			
PROGRAM	O60011_END_PLATE_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, 5/16 THK, PER ASTM B209; FACED BOTH SIDES TO .250			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 1 OF 19
		UNITS	INCH	

PEN PLOTTER: GANTRY END PLATE, OP2 OUTER FACE

O60012



HOLE TABLE (INCH, PROGRAM COORDINATES)					
HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	1.7579	.5079	MGN15H	M3 FHCS CSK	MGN15H BLOCK
2	1.7579	1.4921	MGN15H	M3 FHCS CSK	MGN15H BLOCK
3	2.7421	.5079	MGN15H	M3 FHCS CSK	MGN15H BLOCK
4	2.7421	1.4921	MGN15H	M3 FHCS CSK	MGN15H BLOCK
5	5.2579	.5079	MGN15H	M3 FHCS CSK	MGN15H BLOCK
6	5.2579	1.4921	MGN15H	M3 FHCS CSK	MGN15H BLOCK
7	6.2421	.5079	MGN15H	M3 FHCS CSK	MGN15H BLOCK
8	6.2421	1.4921	MGN15H	M3 FHCS CSK	MGN15H BLOCK
9	3.6063	2.9941	4080 BORE	M5 FHCS CSK	BEAM END
10	3.6063	5.3563	4080 BORE	M5 FHCS CSK	BEAM END
11	4.3937	2.9941	4080 BORE	M5 FHCS CSK	BEAM END
12	4.3937	5.3563	4080 BORE	M5 FHCS CSK	BEAM END
13	2.750	3.875		M3x0.5-6H THRU	Y BELT CLAMP BAR
14	2.750	4.625		M3x0.5-6H THRU	Y BELT CLAMP BAR
15	5.250	3.875		M3x0.5-6H THRU	Y BELT CLAMP BAR
16	5.250	4.625		M3x0.5-6H THRU	Y BELT CLAMP BAR
17	3.500	6.375		M5 CLEARANCE	X BELT TENSION SCREW
18	4.500	6.375		M5 CLEARANCE	X BELT TENSION SCREW
19	2.500	5.500		M3x0.5-6H THRU	CABLE-CHAIN BRACKET
20	2.500	6.250		M3x0.5-6H THRU	CABLE-CHAIN BRACKET
21	5.500	5.500		M3x0.5-6H THRU	CABLE-CHAIN BRACKET
22	5.500	6.250		M3x0.5-6H THRU	CABLE-CHAIN BRACKET

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.

HOLES 1-8: MGN15H BLOCK, 25 MM [.9843] SQ, CENTERED ON X 2.250, Y 1.000 AND X 5.750, Y 1.000.

HOLES 9-12: 4080 BEAM END BORES 20 MM [.7874] X 60 MM [2.3622], CENTERED ON X 4.000; TOP ROW 10 MM [.3937] BELOW THE BEAM TOP (Y 5.750).

FEATURE CALLOUTS

FACE (T16)

FACE THE OUTER FACE (COUNTERSINK SIDE)

STOCK .280, FACE TO .250 ±.005

T16: 1 ROUGH PASS + .005 FINISH

HOLES 13-16

4X M3x0.5-6H THRU

TAP DRILL Ø.0984

Ø.007 Ⓟ .310 A B C

HOLES 17, 18

2X Ø.2165 ±.0040/-0.0010 THRU

M5 CLEARANCE

Ø.009 Ⓜ A B C

HOLES 19-22

4X M3x0.5-6H THRU

TAP DRILL Ø.0984

Ø.004 Ⓟ .250 A B C

HOLES 1-8

8X Ø.1339 ±.0040/-0.0010 THRU

V Ø.2598 ±.0050 X 90°

M3 FLAT HEAD (ISO 10642)

Ø.007 A B C

HOLES 9-12

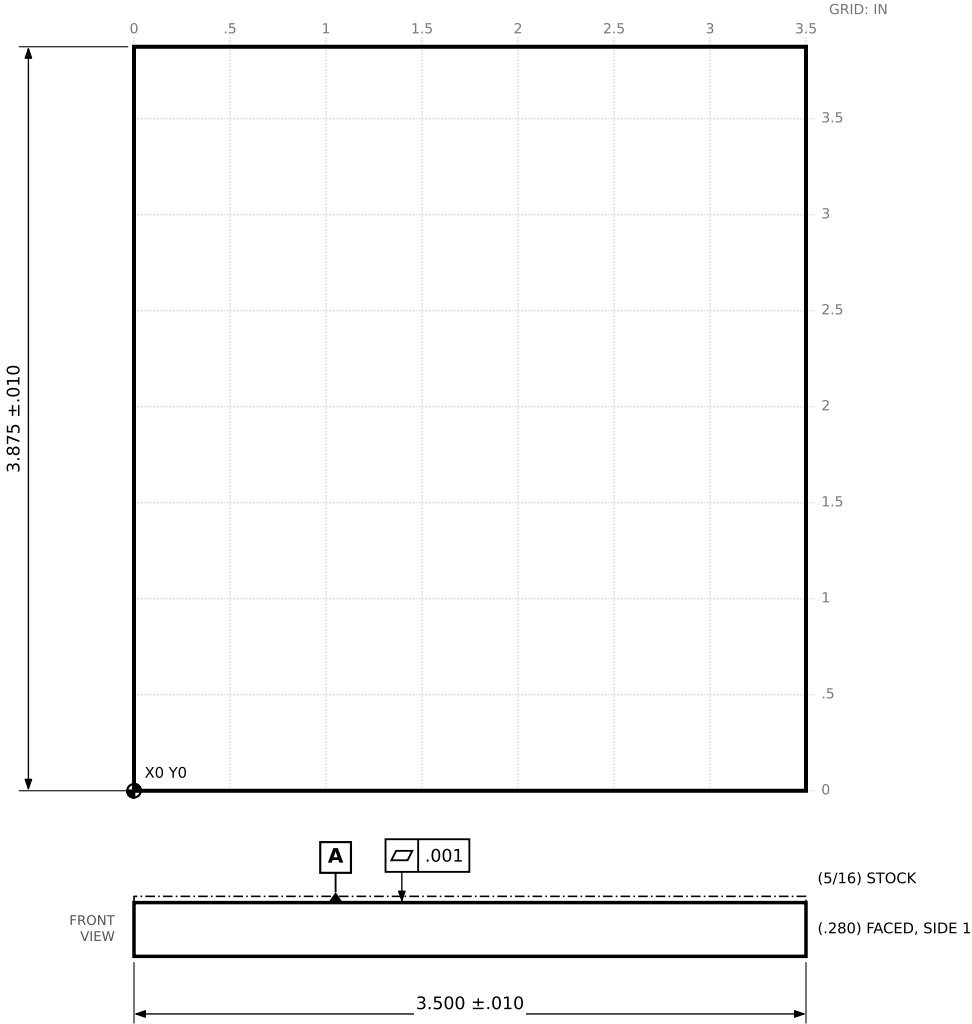
4X Ø.2165 ±.0040/-0.0010 THRU

V Ø.4173 ±.0050 X 90°

M5 FLAT HEAD (ISO 10642)

Ø.009 A B C

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: .XX ±.01 .XXX ±.005 .XXXX ±.0005 ANGLES ±0.5° BREAK ALL EDGES .005-.015, REMOVE ALL BURRS MACHINED SURFACES 125 µin Ra MAX METRIC THREADS PER ASME B1.13M INTERPRET PER ASME Y14.5-2018	NOTES 1. FIXTURE: SIDE 1 (DATUM A) DOWN ON PARALLELS, VISE JAWS PARALLEL TO X ON THE LONG EDGES, FACE UP. 2. ORIENT: LONG SIDE 8.000 ALONG X. 3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .250 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS. 4. DATUMS: A = INNER FACE, AGAINST THE MGN15H BLOCKS (ON THE PARALLELS); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0. 5. COUNTERSINKS FACE OUTWARD (AWAY FROM THE RAIL). SAME PART BOTH SIDES.	PART GANTRY END PLATE			
		PART NO. END_PLATE			
		SETUP OP2 OUTER FACE			
		PROGRAM O60012_END_PLATE_OP2.nc			
		MATERIAL ALUMINUM 6061-T651 PLATE, 5/16 THK, PER ASTM B209; FACED BOTH SIDES TO .250			
		FINISH AS MACHINED			
		QTY 2		SCALE 1:2	SHEET 2 OF 19



NO HOLES IN THIS SETUP: FACING ONLY

- FEATURE CALLOUTS**
- FACE (T16)**
- FACE THE BACK FACE (DATUM A)
 - STOCK 5/16, FACE TO (.280) REF
 - OP2 FACES TO .250 ±.005
 - T16: 1 ROUGH PASS + .005 FINISH

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 μin Ra MAX

METRIC THREADS PER ASME B1.13M

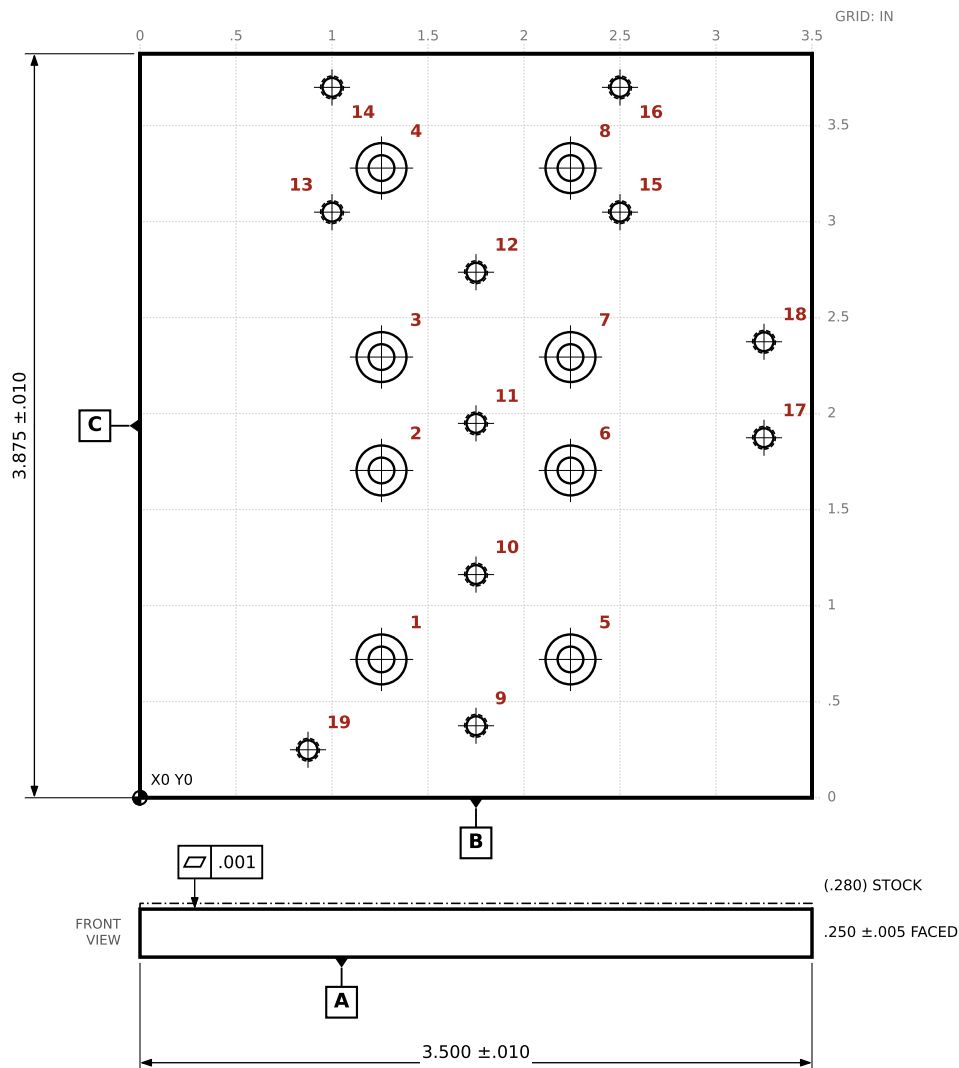
INTERPRET PER ASME Y14.5-2018

- NOTES**
1. FIXTURE: 5/16 PLATE BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, 3.875 APART, BACK FACE UP.
 2. ORIENT: LONG SIDE 3.875 ALONG Y.
 3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .280 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS.
 4. DATUMS: A = BACK FACE, AGAINST THE TWO X MGN15H BLOCKS (FACED IN OP1, UP); B = EDGE AT PROGRAM Y0 IN OP2; C = EDGE AT PROGRAM X0 IN OP2.

PART	PEN CARRIAGE PLATE			
PART NO.	PEN_CARRIAGE			
SETUP	OP1 FACE SIDE 1 (BACK FACE)			
PROGRAM	O60021_PEN_CARRIAGE_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, 5/16 THK, PER ASTM B209; FACED BOTH SIDES TO .250			
FINISH	AS MACHINED			
QTY	1	SCALE	1:1	SHEET 3 OF 19
		UNITS	INCH	

PEN PLOTTER: PEN CARRIAGE PLATE, OP2 FRONT FACE

O60022



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	1.2579	.7205	MGN15H	M3 FHCS CSK	MGN15H BLOCK
2	1.2579	1.7047	MGN15H	M3 FHCS CSK	MGN15H BLOCK
3	1.2579	2.2953	MGN15H	M3 FHCS CSK	MGN15H BLOCK
4	1.2579	3.2795	MGN15H	M3 FHCS CSK	MGN15H BLOCK
5	2.2421	.7205	MGN15H	M3 FHCS CSK	MGN15H BLOCK
6	2.2421	1.7047	MGN15H	M3 FHCS CSK	MGN15H BLOCK
7	2.2421	2.2953	MGN15H	M3 FHCS CSK	MGN15H BLOCK
8	2.2421	3.2795	MGN15H	M3 FHCS CSK	MGN15H BLOCK
9	1.750	.375	MGN9 RAIL	M3x0.5-6H THRU	MGN9 RAIL
10	1.750	1.1624	MGN9 RAIL	M3x0.5-6H THRU	MGN9 RAIL
11	1.750	1.9498	MGN9 RAIL	M3x0.5-6H THRU	MGN9 RAIL
12	1.750	2.7372	MGN9 RAIL	M3x0.5-6H THRU	MGN9 RAIL
13	1.000	3.050		M3x0.5-6H THRU	X BELT CLAMP BAR
14	1.000	3.700		M3x0.5-6H THRU	X BELT CLAMP BAR
15	2.500	3.050		M3x0.5-6H THRU	X BELT CLAMP BAR
16	2.500	3.700		M3x0.5-6H THRU	X BELT CLAMP BAR
17	3.250	1.875		M3x0.5-6H THRU	SERVO BRACKET
18	3.250	2.375		M3x0.5-6H THRU	SERVO BRACKET
19	.875	.250		M3x0.5-6H THRU	DOCK STOP

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.

HOLES 1-8: MGN15H BLOCKS, 25 MM [.9843] SQ EACH ON TWO X RAILS 40 MM [1.5748] APART, CENTERED ON X 1.750, Y 2.000.

HOLES 9-12: MGN9 RAIL HOLES, 20 MM [.7874] PITCH FROM THE FIRST HOLE AT X 1.750, Y .375.

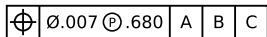
FEATURE CALLOUTS

FACE (T16)

FACE THE FRONT FACE (RAIL SEAT)
STOCK .280, FACE TO .250 ±.005
T16: 1 ROUGH PASS + .005 FINISH

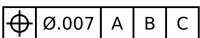
HOLES 13-16

4X M3x0.5-6H THRU
TAP DRILL Ø.0984



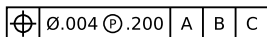
HOLES 1-8

8X Ø.1339 +.0040/-.0010 THRU
V Ø.2598 ±.0050 X 90°
M3 FLAT HEAD (ISO 10642)



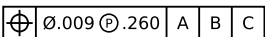
HOLES 17, 18

2X M3x0.5-6H THRU
TAP DRILL Ø.0984



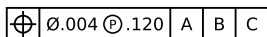
HOLES 9-12

4X M3x0.5-6H THRU
TAP DRILL Ø.0984



HOLE 19

M3x0.5-6H THRU
TAP DRILL Ø.0984



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

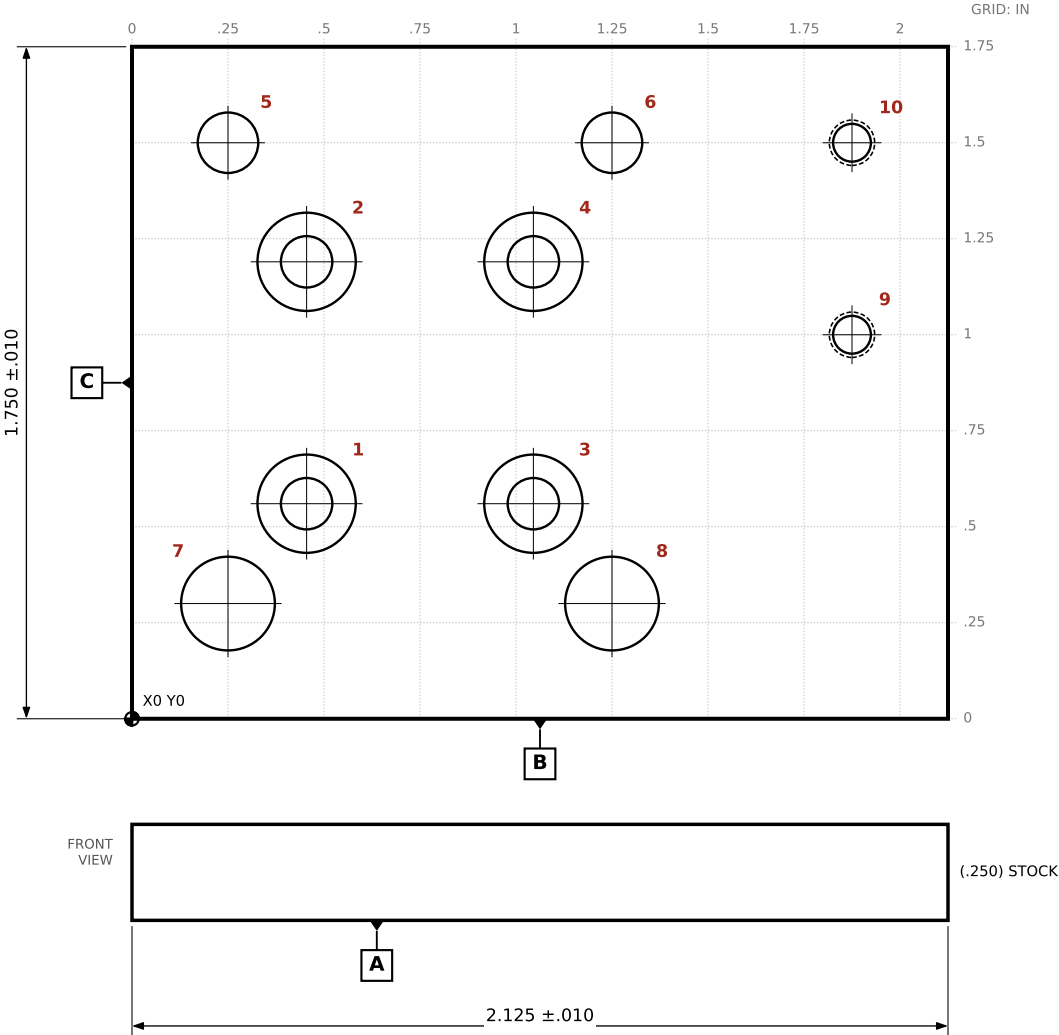
NOTES

1. FIXTURE: SIDE 1 (DATUM A) DOWN ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, 3.875 APART, FACE UP.
2. ORIENT: LONG SIDE 3.875 ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .250 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS.
4. DATUMS: A = BACK FACE, AGAINST THE TWO X MGN15H BLOCKS (ON THE PARALLELS); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.
5. COUNTERSINKS FACE FORWARD (AWAY FROM THE BEAM).

PART	PEN CARRIAGE PLATE			
PART NO.	PEN_CARRIAGE			
SETUP	OP2 FRONT FACE			
PROGRAM	O60022_PEN_CARRIAGE_OP2.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, 5/16 THK, PER ASTM B209; FACED BOTH SIDES TO .250			
FINISH	AS MACHINED			
QTY	1	SCALE	1:1	SHEET 4 OF 19
		UNITS	INCH	

PEN PLOTTER: PEN DOCK, OP1 FRONT FACE

O60031



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.4547	.5600	MGN9H	M3 SHCS CBORE	MGN9H BLOCK, M3X5 SHCS
2	.4547	1.1900	MGN9H	M3 SHCS CBORE	MGN9H BLOCK, M3X5 SHCS
3	1.0453	.5600	MGN9H	M3 SHCS CBORE	MGN9H BLOCK, M3X5 SHCS
4	1.0453	1.1900	MGN9H	M3 SHCS CBORE	MGN9H BLOCK, M3X5 SHCS
5	.250	1.500		Ø4 MM DOWEL REAM	4 MM DOWEL, PRESS FIT + RETAINING COMPOUND
6	1.250	1.500		Ø4 MM DOWEL REAM	4 MM DOWEL, PRESS FIT + RETAINING COMPOUND
7	.250	.300		MAGNET POCKET	6 X 3 MM N52 MAGNET, EPOXY
8	1.250	.300		MAGNET POCKET	6 X 3 MM N52 MAGNET, EPOXY
9	1.875	1.000		M3X0.5-6H THRU	LIFT TAB (M3X10 SHCS, HEAD FORWARD)
10	1.875	1.500		M3X0.5-6H THRU	LIFT TAB (M3X10 SHCS, HEAD FORWARD)

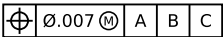
X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.
HOLES 1-4: MGN9H BLOCK, 15 MM [.5906] X 16 MM [.6299], CENTERED ON X .750, Y .875.

FEATURE CALLOUTS

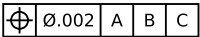
HOLES 1-4

4X Ø.1339 +.0040/-.0010 THRU
LJ Ø.2559 ±.0050 ±.1575 ±.0050
M3 SOCKET HEAD (ISO 4762)



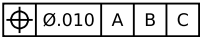
HOLES 5, 6

2X Ø.1575-.1579 THRU, REAM
Ø4 MM H7 FOR MISUMI MSC4-12 DOWEL



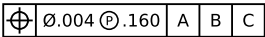
HOLES 7, 8

2X Ø.2441 +.0040/-.0000 ±.1260 ±.0050
FLAT BOTTOM, FOR Ø6 X 3 MM MAGNET



HOLES 9, 10

2X M3X0.5-6H THRU
TAP DRILL Ø.0984



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

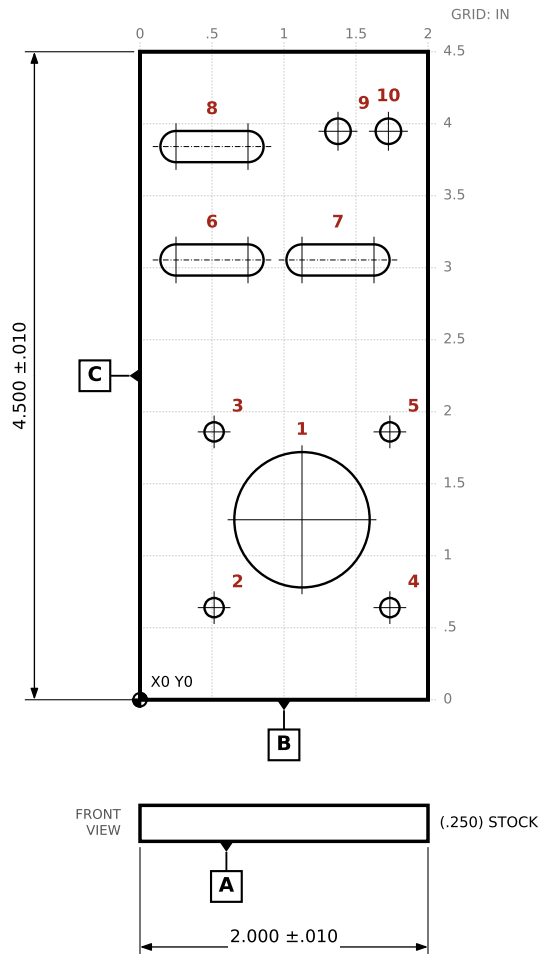
NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE LONG EDGES, FACE UP.
2. ORIENT: LONG SIDE 2.125 ALONG X.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = BACK FACE, AGAINST THE MGN9H BLOCK (ON THE PARALLELS); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.

PART	PEN DOCK			
PART NO.	PEN_DOCK			
SETUP	OP1 FRONT FACE			
PROGRAM	O60031_PEN_DOCK_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	1	SCALE	2:1	SHEET 5 OF 19
		UNITS	INCH	

PEN PLOTTER: X MOTOR / IDLER PLATE, OP1 TOP FACE

O60051



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	1.125	1.250		PILOT CLEARANCE BORE	NEMA17 PILOT / IDLER BOLT CLEARANCE
2	.5148	.6398	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
3	.5148	1.8602	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
4	1.7352	.6398	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
5	1.7352	1.8602	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
6	.500	3.0541	BEAM SLOT	M5 SLOT	BEAM T-NUT, SLIDES FOR X BELT TENSION
7	1.375	3.0541	BEAM SLOT	M5 SLOT	BEAM T-NUT
8	.500	3.8415	BEAM SLOT	M5 SLOT	BEAM T-NUT
9	1.375	3.9478	BEAM CL	M4 CLEARANCE	TENSION BLOCK (IDLER SIDE)
10	1.725	3.9478	BEAM CL	M4 CLEARANCE	TENSION BLOCK (IDLER SIDE)

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.

HOLES 2-5: NEMA 17 SCREWS, 31 MM [1.2205] SQ, CENTERED ON X 1.125, Y 1.250.

HOLES 6-8: ON THE BEAM TOP T-SLOTS, 10 MM [.3937] EACH SIDE OF THE BEAM CENTERLINE AT Y 3.4478.

HOLES 9, 10: .500 FROM THE BEAM CENTERLINE AT Y 3.4478 (TENSION SCREW .500 FROM THE GANTRY CENTERLINE).

FEATURE CALLOUTS

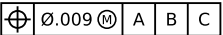
HOLE 1

Ø.940 +.005/-.000 THRU
NEMA 17 PILOT CLEARANCE



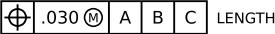
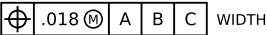
HOLES 2-5

4X Ø.1339 +.0040/-.0010 THRU
M3 CLEARANCE



HOLES 6-8

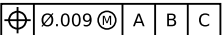
3X SLOT .2165 +.0040/-.0010 WIDE
X .7165 ±.0100 LONG, THRU
FULL R ENDS, M5 CLEARANCE



BOUNDARY

HOLES 9, 10

2X Ø.1772 +.0040/-.0010 THRU
M4 CLEARANCE



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

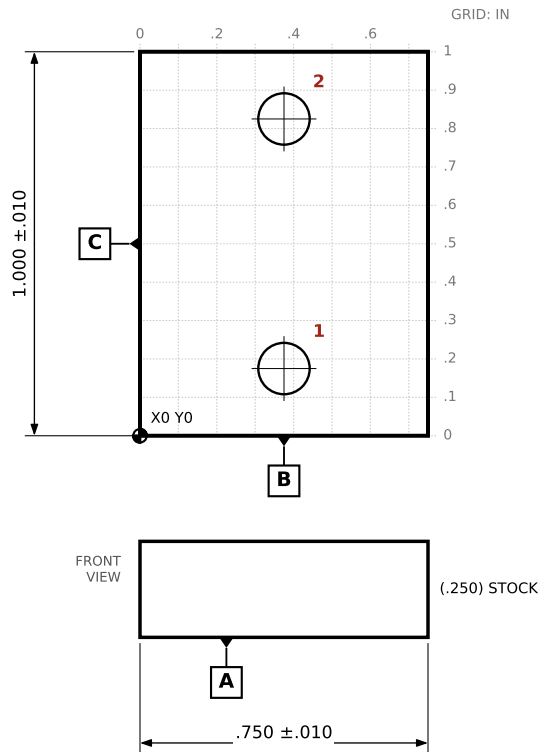
METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, 4.500 APART, FACE UP.
2. ORIENT: LONG SIDE 4.500 ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = FACE ON THE PARALLELS (MOTOR SIDE: SITS ON THE BEAM TOP); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.
5. RIGHT SIDE CARRIES THE X MOTOR; THE LEFT ONE IS THE SAME PART FLIPPED OVER AND CARRIES THE IDLER.

PART	X MOTOR / IDLER PLATE			
PART NO.	X_MOTOR_PLATE			
SETUP	OP1 TOP FACE			
PROGRAM	O60051_X_MOTOR_PLATE_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	2	SCALE	3:4	SHEET 6 OF 19
		UNITS	INCH	



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.375	.175		M3 CLEARANCE	BELT CLAMP SCREW
2	.375	.825		M3 CLEARANCE	BELT CLAMP SCREW

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

HOLES 1, 2

2X Ø.1339 +.0040/-.0010 THRU
M3 CLEARANCE

⊕	Ø.007	M	A	B	C
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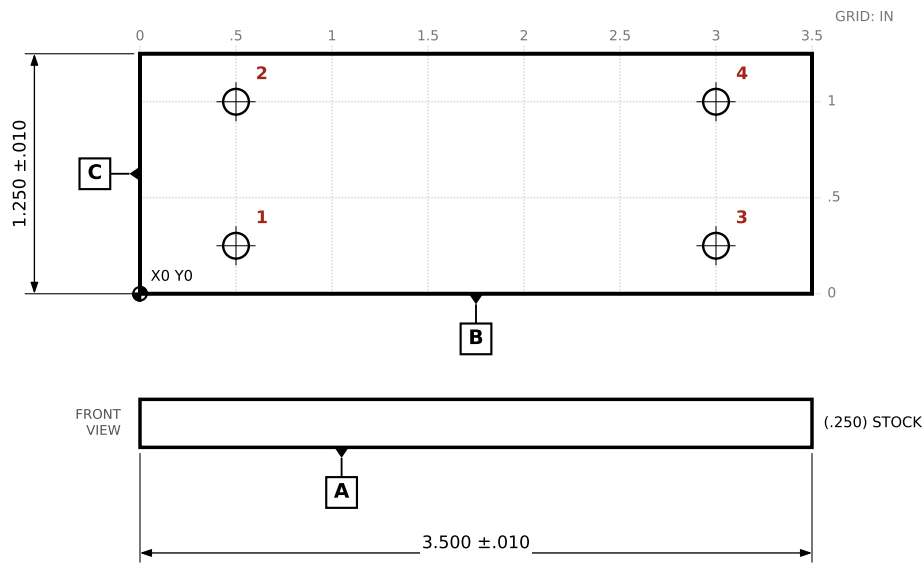
UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
.XX ±.01 .XXX ±.005 .XXXX ±.0005
ANGLES ±0.5°
BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 µin Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, 1.000 APART, FACE UP.
2. ORIENT: LONG SIDE 1.000 ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = FACE ON THE PARALLELS (OPPOSITE THE MACHINED FACE); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.

PART	X BELT CLAMP BAR			
PART NO.	X_CLAMP_BAR			
SETUP	OP1			
PROGRAM	O60061_X_CLAMP_BAR_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	2	SCALE	2:1	SHEET 7 OF 19
		UNITS	INCH	



HOLE TABLE (INCH, PROGRAM COORDINATES)

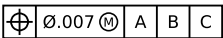
HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.500	.250		M3 CLEARANCE	BELT CLAMP SCREW
2	.500	1.000		M3 CLEARANCE	BELT CLAMP SCREW
3	3.000	.250		M3 CLEARANCE	BELT CLAMP SCREW
4	3.000	1.000		M3 CLEARANCE	BELT CLAMP SCREW

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

HOLES 1-4

4X Ø.1339 +.0040/-.0010 THRU
M3 CLEARANCE



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

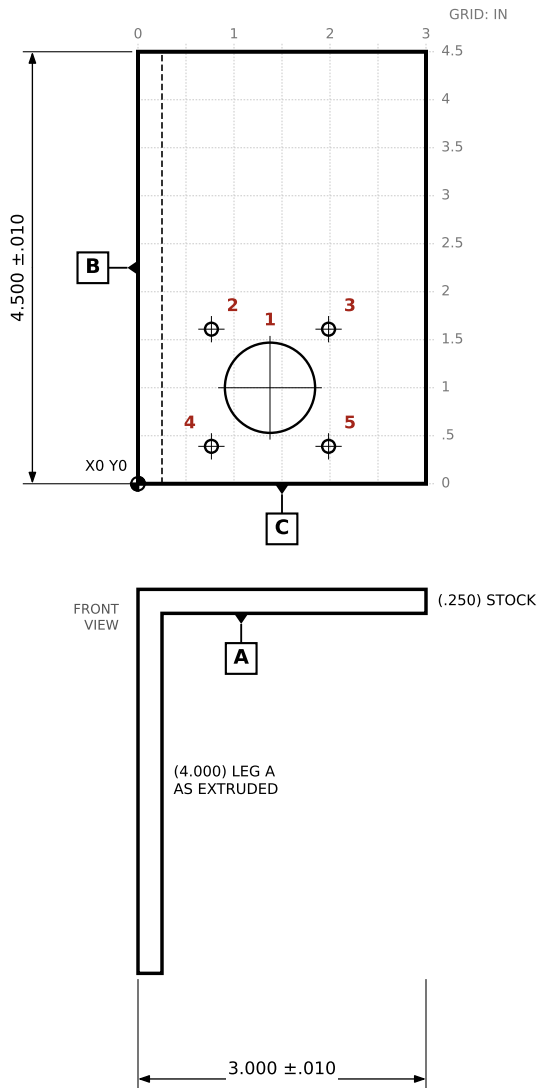
NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE LONG EDGES, FACE UP.
2. ORIENT: LONG SIDE 3.500 ALONG X.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = FACE ON THE PARALLELS (OPPOSITE THE MACHINED FACE); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.

PART	Y BELT CLAMP BAR		
PART NO.	Y_CLAMP_BAR		
SETUP	OP1		
PROGRAM	O60071_Y_CLAMP_BAR_OP1.nc		
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209		
FINISH	AS MACHINED		
QTY	2	SCALE	1:1
SHEET	8 OF 19	UNITS	INCH

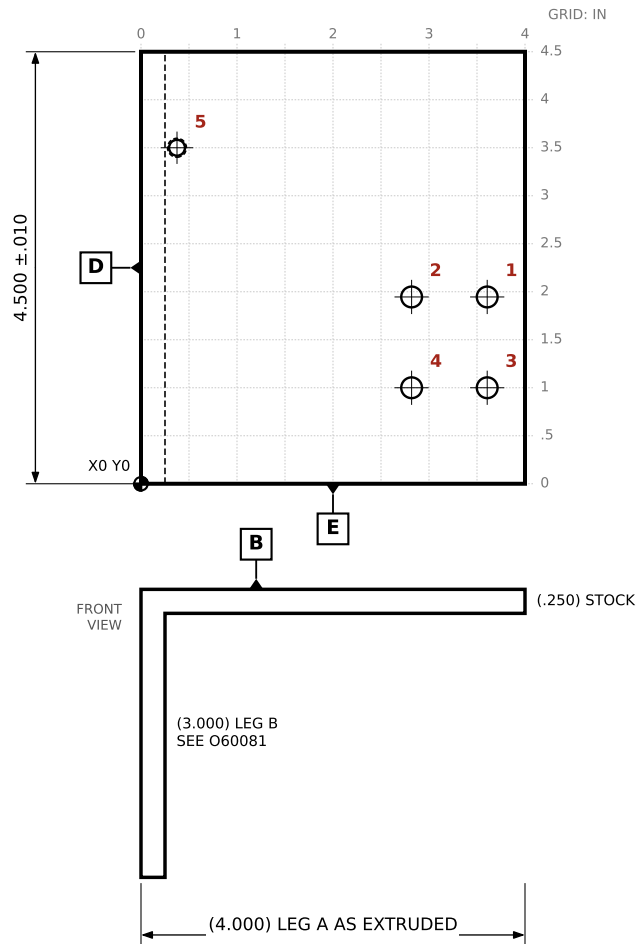
PEN PLOTTER: Y MOTOR/IDLER BRACKET A, OP1 LEG B (HORIZONTAL LEG, TOP FACE)

060081



PEN PLOTTER: Y MOTOR/IDLER BRACKET A, OP2 LEG A (VERTICAL LEG, OUTER FACE)

060082



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	3.6063	1.9459	2040 BORE	M5 CLEARANCE	FRAME END TAP
2	2.8189	1.9459	2040 BORE	M5 CLEARANCE	FRAME END TAP
3	3.6063	1.000	2040 SLOT	M5 CLEARANCE	CROSS-MEMBER T-NUT
4	2.8189	1.000	2040 SLOT	M5 CLEARANCE	CROSS-MEMBER T-NUT
5	.375	3.500		M5x0.8-6H THRU	IDLER-CARRIER JACK SCREW (DRILL THROUGH FILLET)

X FROM DATUM D, Y FROM DATUM E. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.

HOLES 1, 2: 2040 END BORES ON THE SIDE MEMBER CENTERLINE, 10 MM [.3937] AND 30 MM [1.1811] FROM LEG A'S FREE EDGE.

HOLES 3, 4: 2040 T-SLOTS, 10 MM [.3937] AND 30 MM [1.1811] FROM LEG A'S FREE EDGE.

FEATURE CALLOUTS

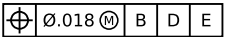
HOLES 1, 2

2X Ø.2165 +.0040/-.0010 THRU
M5 CLEARANCE



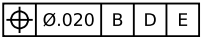
HOLES 3, 4

2X Ø.2165 +.0040/-.0010 THRU
M5 CLEARANCE



HOLE 5

M5x0.8-6H THRU
TAP DRILL Ø.1654



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

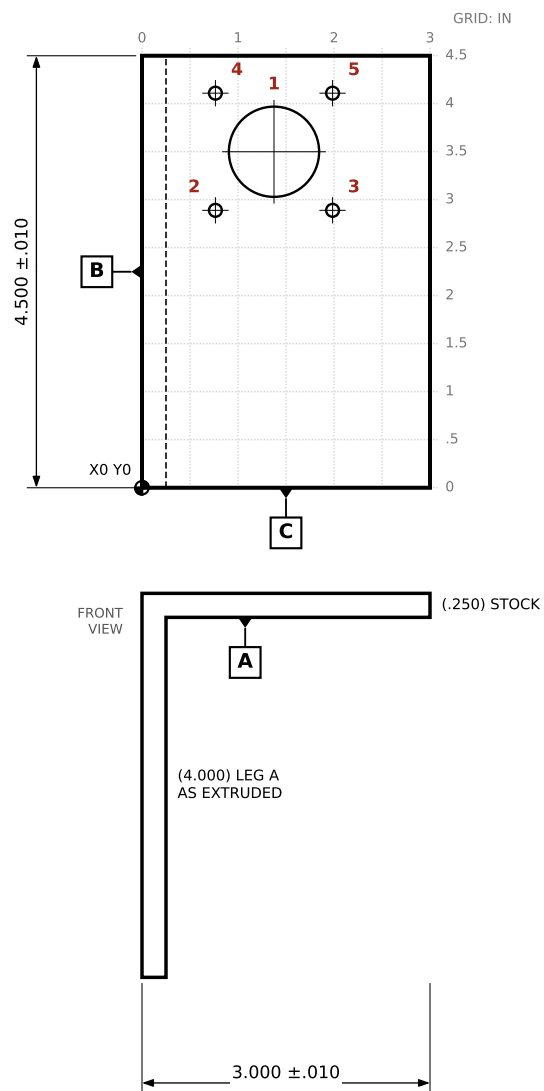
NOTES

1. FIXTURE: GRIP BY THE SAWN ENDS. LEG A FLAT ON PARALLELS, FRAME-SIDE FACE UP.
2. ORIENT: LONG SIDE 4.500 ALONG Y. CORNER AT X0 - LEG B HANGS DOWN PAST THE X-END OF THE JAWS.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = LEG B INNER FACE (MOTOR SEAT, ON THE PARALLELS IN OP1); B = LEG A OUTER FACE (FRAME SIDE); C = SAWN END AT PROGRAM Y0 IN OP1; D = LEG B OUTER FACE (PROGRAM X0 IN OP2); E = OPPOSITE SAWN END (PROGRAM Y0 IN OP2).
5. BOLTS TO THE FRAME END; DOUBLES AS THE CORNER JOINT.

PART	Y MOTOR/IDLER BRACKET A			
PART NO.	Y_BRACKET_A			
SETUP	OP2 LEG A (VERTICAL LEG, OUTER FACE)			
PROGRAM	060082_Y_BRACKET_A_OP2.nc			
MATERIAL	ALUMINUM 6061-T6 EXTRUDED ANGLE 4 X 4 X .250, PER ASTM B221			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 10 OF 19
		UNITS	INCH	

PEN PLOTTER: Y MOTOR/IDLER BRACKET B (MIRROR OF A), OP1 LEG B (HORIZONTAL LEG, TOP FACE)

060091



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	1.375	3.500		PILOT CLEARANCE BORE	NEMA17 PILOT (MOTOR END) / IDLER BOLT CLEARANCE (IDLER END)
2	.7648	2.8898	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
3	1.9852	2.8898	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
4	.7648	4.1102	NEMA17	M3 CLEARANCE	NEMA17 SCREWS
5	1.9852	4.1102	NEMA17	M3 CLEARANCE	NEMA17 SCREWS

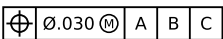
X FROM DATUM B, Y FROM DATUM C. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.
HOLES 2-5: NEMA 17 SCREWS, 31 MM [1.2205] SQ, CENTERED ON X 1.375, Y 3.500.

FEATURE CALLOUTS

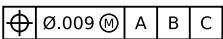
HOLE 1

Ø.940 +.005/-.000 THRU
NEMA 17 PILOT CLEARANCE



HOLES 2-5

4X Ø.1339 +.0040/-.0010 THRU
M3 CLEARANCE



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005
ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 µin Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

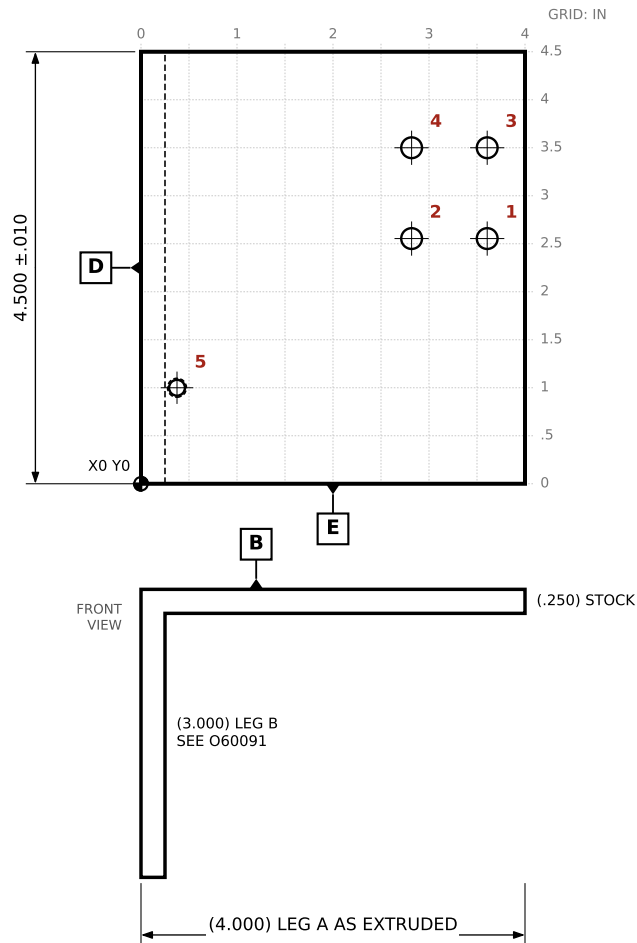
NOTES

1. FIXTURE: GRIP BY THE TWO SAWN ENDS - 4.500 APART ALONG Y. LEG B FLAT ON PARALLELS, OUTER FACE UP.
2. ORIENT: LONG SIDE 4.500 ALONG Y. CORNER AT X0 - LEG A HANGS DOWN PAST THE X-END OF THE JAWS.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = LEG B INNER FACE (MOTOR SEAT, ON THE PARALLELS IN OP1); B = LEG A OUTER FACE (FRAME SIDE); C = SAWN END AT PROGRAM Y0 IN OP1; D = LEG B OUTER FACE (PROGRAM X0 IN OP2); E = OPPOSITE SAWN END (PROGRAM Y0 IN OP2).
5. BOLTS TO THE FRAME END; DOUBLES AS THE CORNER JOINT.

PART	Y MOTOR/IDLER BRACKET B (MIRROR OF A)			
PART NO.	Y_BRACKET_B			
SETUP	OP1 LEG B (HORIZONTAL LEG, TOP FACE)			
PROGRAM	060091_Y_BRACKET_B_OP1.nc			
MATERIAL	ALUMINUM 6061-T6 EXTRUDED ANGLE 4 X 4 X .250, PER ASTM B221			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 11 OF 19 UNITS INCH

PEN PLOTTER: Y MOTOR/IDLER BRACKET B (MIRROR OF A), OP2 LEG A (VERTICAL LEG, OUTER FACE)

060092



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	3.6063	2.5541	2040 BORE	M5 CLEARANCE	FRAME END TAP
2	2.8189	2.5541	2040 BORE	M5 CLEARANCE	FRAME END TAP
3	3.6063	3.500	2040 SLOT	M5 CLEARANCE	CROSS-MEMBER T-NUT
4	2.8189	3.500	2040 SLOT	M5 CLEARANCE	CROSS-MEMBER T-NUT
5	.375	1.000		M5x0.8-6H THRU	IDLER-CARRIER JACK SCREW (DRILL THROUGH FILLET)

X FROM DATUM D, Y FROM DATUM E. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.
HOLES 1, 2: 2040 END BORES ON THE SIDE MEMBER CENTERLINE, 10 MM [.3937] AND 30 MM [1.1811] FROM LEG A'S FREE EDGE.

HOLES 3, 4: 2040 T-SLOTS, 10 MM [.3937] AND 30 MM [1.1811] FROM LEG A'S FREE EDGE.

FEATURE CALLOUTS

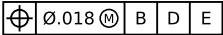
HOLES 1, 2

2X Ø.2165 +.0040/-.0010 THRU
M5 CLEARANCE



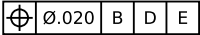
HOLES 3, 4

2X Ø.2165 +.0040/-.0010 THRU
M5 CLEARANCE



HOLE 5

M5x0.8-6H THRU
TAP DRILL Ø.1654



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

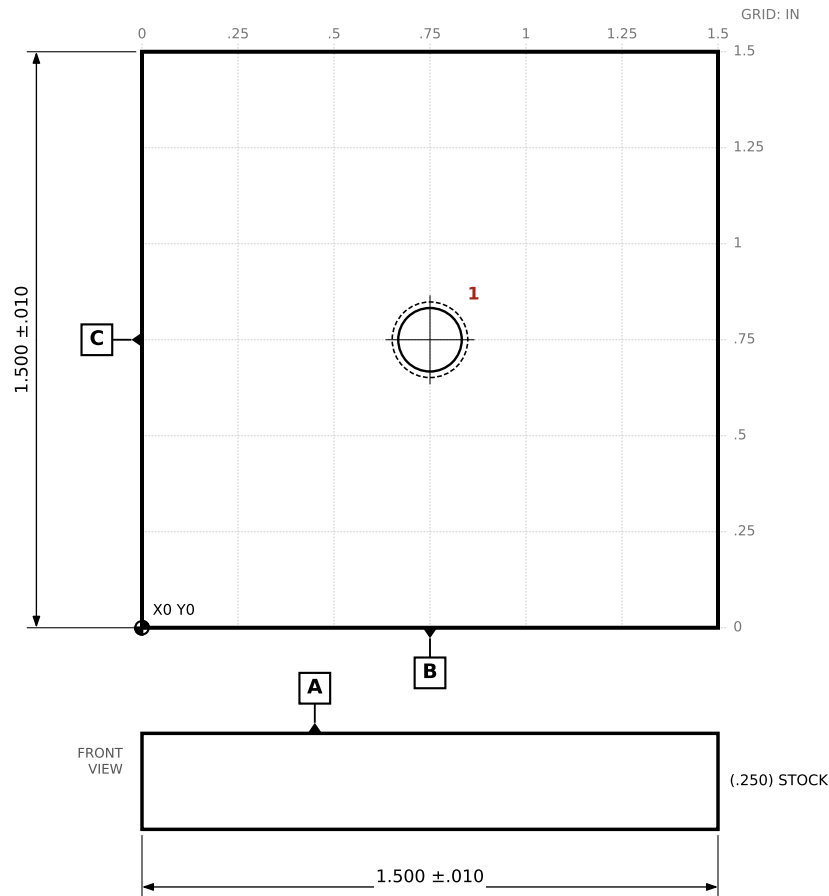
METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: GRIP BY THE SAWN ENDS. LEG A FLAT ON PARALLELS, FRAME-SIDE FACE UP.
2. ORIENT: LONG SIDE 4.500 ALONG Y. CORNER AT X0 - LEG B HANGS DOWN PAST THE X-END OF THE JAWS.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = LEG B INNER FACE (MOTOR SEAT, ON THE PARALLELS IN OP1); B = LEG A OUTER FACE (FRAME SIDE); C = SAWN END AT PROGRAM Y0 IN OP1; D = LEG B OUTER FACE (PROGRAM X0 IN OP2); E = OPPOSITE SAWN END (PROGRAM Y0 IN OP2).
5. BOLTS TO THE FRAME END; DOUBLES AS THE CORNER JOINT.

PART	Y MOTOR/IDLER BRACKET B (MIRROR OF A)			
PART NO.	Y_BRACKET_B			
SETUP	OP2 LEG A (VERTICAL LEG, OUTER FACE)			
PROGRAM	060092_Y_BRACKET_B_OP2.nc			
MATERIAL	ALUMINUM 6061-T6 EXTRUDED ANGLE 4 X 4 X .250, PER ASTM B221			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 12 OF 19
		UNITS	INCH	



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.750	.750		M5x0.8-6H THRU	IDLER AXLE BOLT (M5 SHCS FROM ABOVE)

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

HOLE 1

M5x0.8-6H THRU
TAP DRILL Ø.1654

⊕	Ø.030	A	B	C
⊥	Ø.005	Ⓢ1.000	A	

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

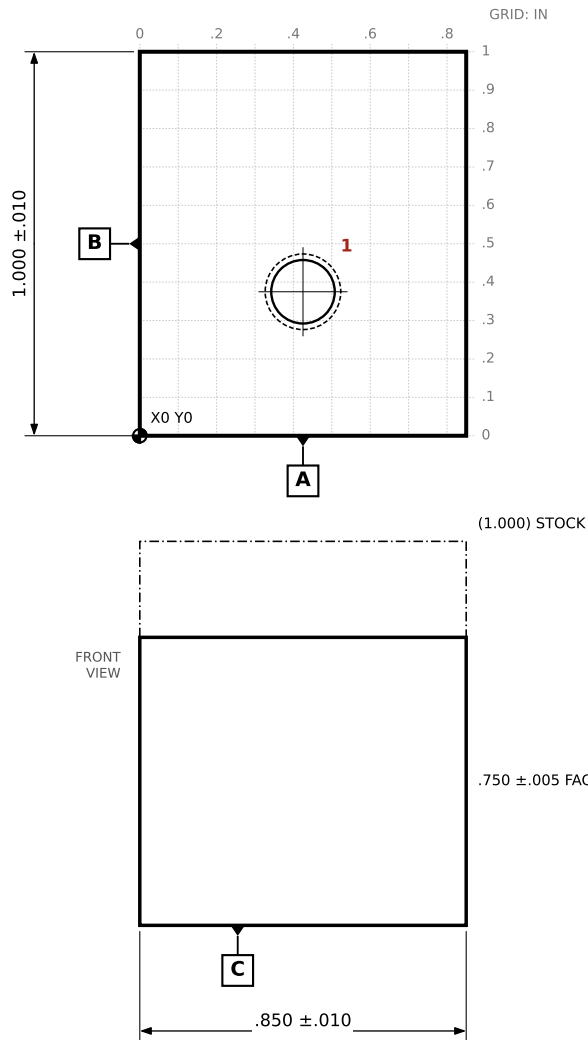
NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X, EITHER PAIR OF EDGES, FACE UP.
2. ORIENT: SQUARE BLANK 1.500 X 1.500.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = MACHINED (TOP) FACE, AGAINST BRACKET LEG B AND THE AXLE SPACER; B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.

PART	Y IDLER CARRIER			
PART NO.	Y_IDLER_CARRIER			
SETUP	OP1			
PROGRAM	O60101_Y_IDLER_CARRIER_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	2	SCALE	2:1	SHEET 13 OF 19
		UNITS	INCH	

PEN PLOTTER: X BELT TENSION BLOCK, OP1 END FACE (FACES THE LEFT END PLATE)

O60111



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.425	.375		M5x0.8-6H BLIND	X BELT TENSION SCREW (THRU LEFT END PLATE)

X FROM DATUM B, Y FROM DATUM A. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

FACE (T16)

FACE THE SIDE OPPOSITE DATUM C
STOCK 1.000, FACE TO .750 ±.005
T16: 5 ROUGH PASSES + .005 FINISH

HOLE 1

M5x0.8-6H $\pm$.530 MIN FULL THREAD
TAP DRILL $\varnothing$.1654

$\varnothing$	.009	$\varnothing$	.450	A	B	C
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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 μ in Ra MAX

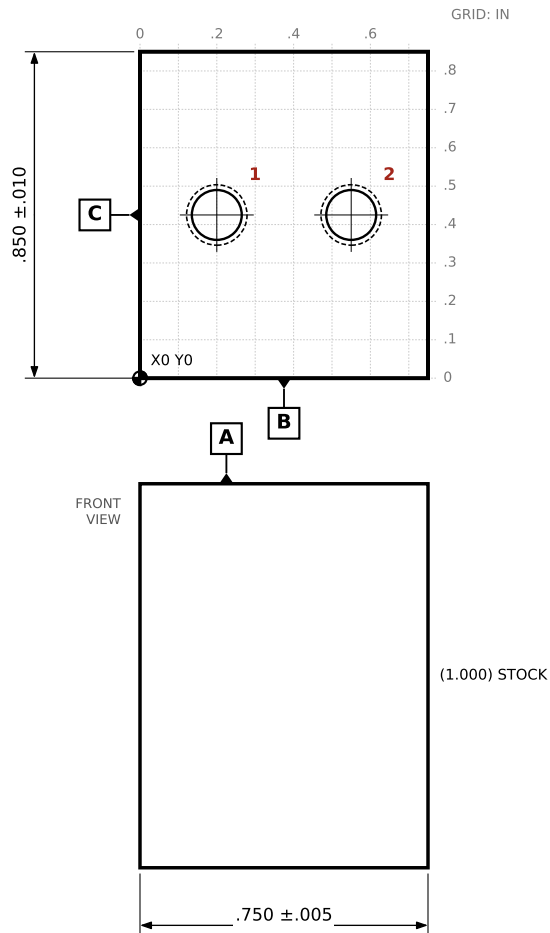
METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: STAND THE BAR ON END IN THE VISE: THE SIDE THAT STAYS AS EXTRUDED (DATUM C) DOWN ON THE PARALLELS, JAWS ON THE BOTTOM FACE AND THE TOP FACE.
2. ORIENT: LONG SIDE 1.000 ALONG Y: X ALONG THE .850 SAWN LENGTH, Y0 = BOTTOM FACE EDGE.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .750 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS.
4. DATUMS: A = BOTTOM FACE (DRILLED IN OP2), SITS ON THE X IDLER PLATE; B = SIDE AT OP2 PROGRAM Y0 (OP1 PROGRAM X0); C = END AT OP2 PROGRAM X0, AS EXTRUDED (ON THE PARALLELS IN OP1).

PART	X BELT TENSION BLOCK		
PART NO.	X_TENSION_BLOCK		
SETUP	OP1 END FACE (FACES THE LEFT END PLATE)		
PROGRAM	O60111_X_TENSION_BLOCK_OP1.nc		
MATERIAL	ALUMINUM 6061-T6511 EXTRUDED SQUARE BAR 1.000 X 1.000, PER ASTM B221		
FINISH	AS MACHINED		
QTY	1	SCALE	2:1
SHEET	14 OF 19	UNITS	INCH



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.200	.425		M4x0.7-6H BLIND	IDLER PLATE SCREW (FROM BELOW)
2	.550	.425		M4x0.7-6H BLIND	IDLER PLATE SCREW (FROM BELOW)

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

HOLES 1, 2
2X M4x0.7-6H $\perp$.290 MIN FULL THREAD
TAP DRILL $\varnothing$.1299

$\varnothing$	$\varnothing$.009	$\varnothing$.250	A	B	C
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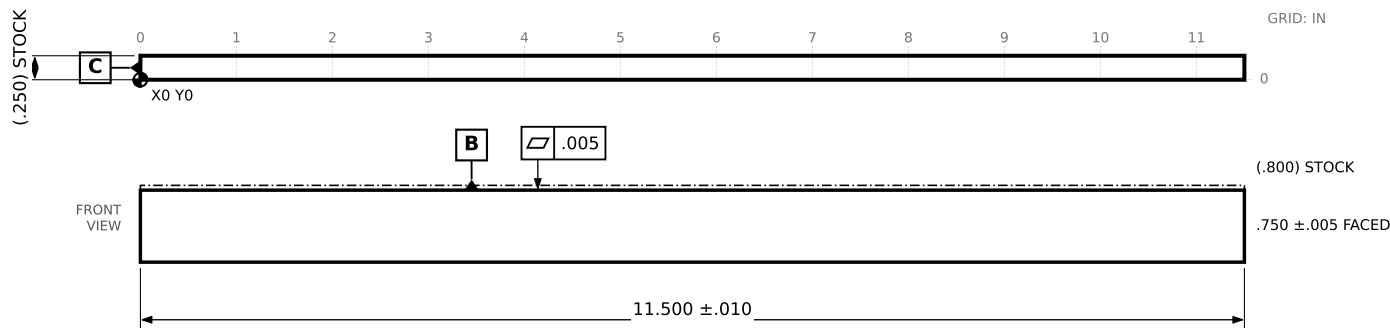
UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.XX ± .01 .XXX ± .005 .XXXX ± .0005
ANGLES ± 0.5°
BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 μ in Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

NOTES
1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, .850 APART, FACE UP.
2. ORIENT: LONG SIDE .850 ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = BOTTOM FACE (DRILLED IN OP2), SITS ON THE X IDLER PLATE; B = SIDE AT OP2 PROGRAM Y0 (OP1 PROGRAM X0); C = END AT OP2 PROGRAM X0, AS EXTRUDED (ON THE PARALLELS IN OP1).

PART	X BELT TENSION BLOCK			
PART NO.	X_TENSION_BLOCK			
SETUP	OP2 BOTTOM FACE			
PROGRAM	O60112_X_TENSION_BLOCK_OP2.nc			
MATERIAL	ALUMINUM 6061-T6511 EXTRUDED SQUARE BAR 1.000 X 1.000, PER ASTM B221			
FINISH	AS MACHINED			
QTY	1	SCALE	2:1	SHEET 15 OF 19 UNITS INCH

PEN PLOTTER: PAPER CORNER STOP BAR, OP1 FACE THE REGISTRATION EDGE

O60121



NO HOLES IN THIS SETUP: FACING ONLY

FEATURE CALLOUTS

FACE (T16)
FACE THE REGISTRATION EDGE (DATUM B)
STOCK .800, FACE TO .750 ± .005
T16: 1 ROUGH PASS + .005 FINISH

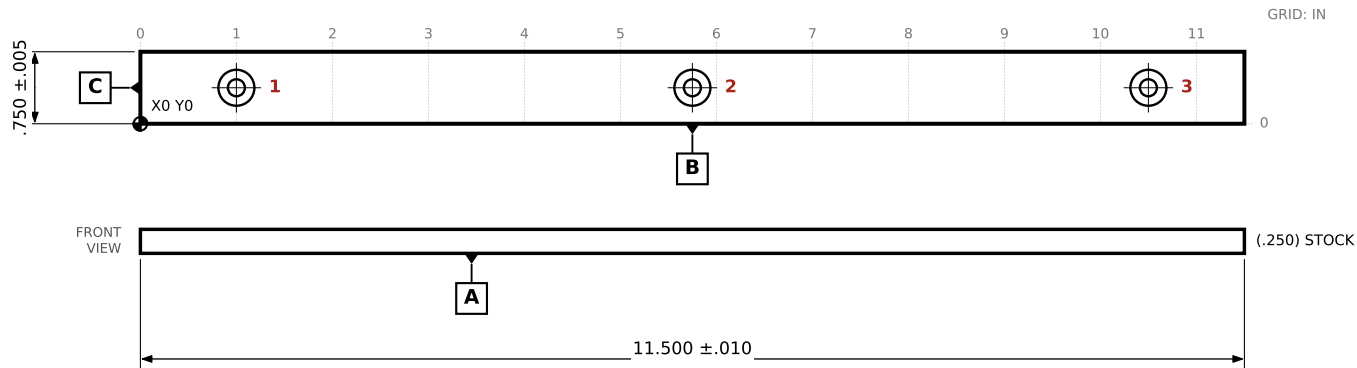
UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
.XX ± .01 .XXX ± .005 .XXXX ± .0005
ANGLES ± 0.5°
BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 µin Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: STAND THE STRIP ON ITS SAWN EDGE ON PARALLELS, REGISTRATION EDGE UP, CLAMPED BETWEEN THE FIXED JAW AND A 12 IN PARALLEL SO THE ENDS ARE SUPPORTED.
2. ORIENT: LONG SIDE 11.500 ALONG X, THE .250 THICKNESS ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = FINISHED FACE, SET .750 ABOVE THE PARALLELS. TOP OF THE JAWS BELOW Z0: THE FACE MILL OVERHANGS THE JAWS.
4. DATUMS: A = FACE ON THE PARALLELS IN OP2, SITS ON THE BED; B = PAPER-REGISTRATION EDGE (FACED IN OP1, UP; PROGRAM Y0 IN OP2); C = END AT PROGRAM X0.

PART	PAPER CORNER STOP BAR			
PART NO.	PAPER_STOP			
SETUP	OP1 FACE THE REGISTRATION EDGE			
PROGRAM	O60121_PAPER_STOP_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209; STRIP SAWN .800 WIDE			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 16 OF 19
		UNITS	INCH	



FEATURE CALLOUTS

HOLES 1-3
3X Ø.1772 +.0040/- .0010 THRU
V Ø.375 ±.010 X 90°
#8 FLAT HEAD WOOD SCREW

	Ø.030	A	B	C
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HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	1.000	.375		#8 WOOD SCREW CSK	#8 FH WOOD SCREW TO THE BED
2	5.750	.375		#8 WOOD SCREW CSK	#8 FH WOOD SCREW TO THE BED
3	10.500	.375		#8 WOOD SCREW CSK	#8 FH WOOD SCREW TO THE BED

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

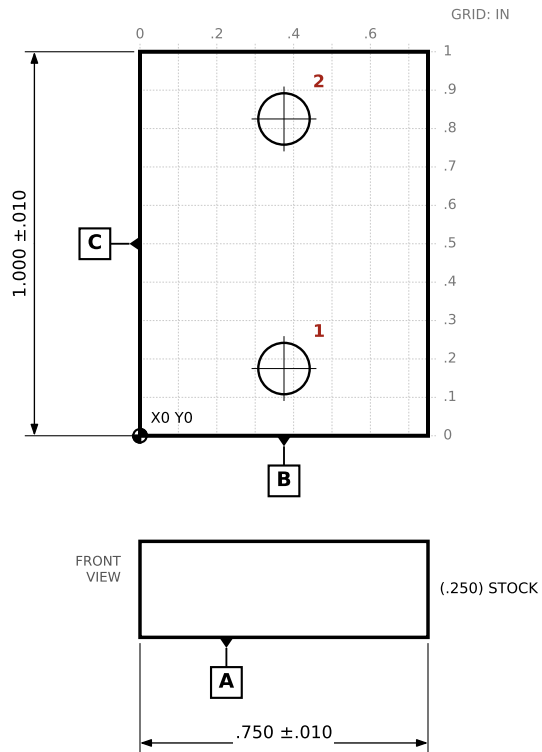
UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
.XX ±.01 .XXX ±.005 .XXXX ±.0005
ANGLES ±0.5°
BREAK ALL EDGES .005-.015, REMOVE ALL BURRS
MACHINED SURFACES 125 µin Ra MAX
METRIC THREADS PER ASME B1.13M
INTERPRET PER ASME Y14.5-2018

NOTES

- 1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE LONG EDGES, FACED EDGE AT Y0, FACE UP.
- 2. ORIENT: LONG SIDE 11.500 ALONG X.
- 3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
- 4. DATUMS: A = FACE ON THE PARALLELS, SITS ON THE BED; B = EDGE AT PROGRAM Y0; THE PAPER-REGISTRATION EDGE (FACED IN OP1); C = EDGE AT PROGRAM X0.

PART	PAPER CORNER STOP BAR			
PART NO.	PAPER_STOP			
SETUP	OP2			
PROGRAM	O60122_PAPER_STOP_OP2.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209; STRIP SAWN .800 WIDE			
FINISH	AS MACHINED			
QTY	2	SCALE	1:2	SHEET 17 OF 19
		UNITS	INCH	



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.375	.175		M3 CLEARANCE	BELT CLAMP SCREW
2	.375	.825		M3 CLEARANCE	BELT CLAMP SCREW

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

FEATURE CALLOUTS

HOLES 1, 2

2X Ø.1339 +.0040/-.0010 THRU
M3 CLEARANCE

Ø.007	M	A	B	C
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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

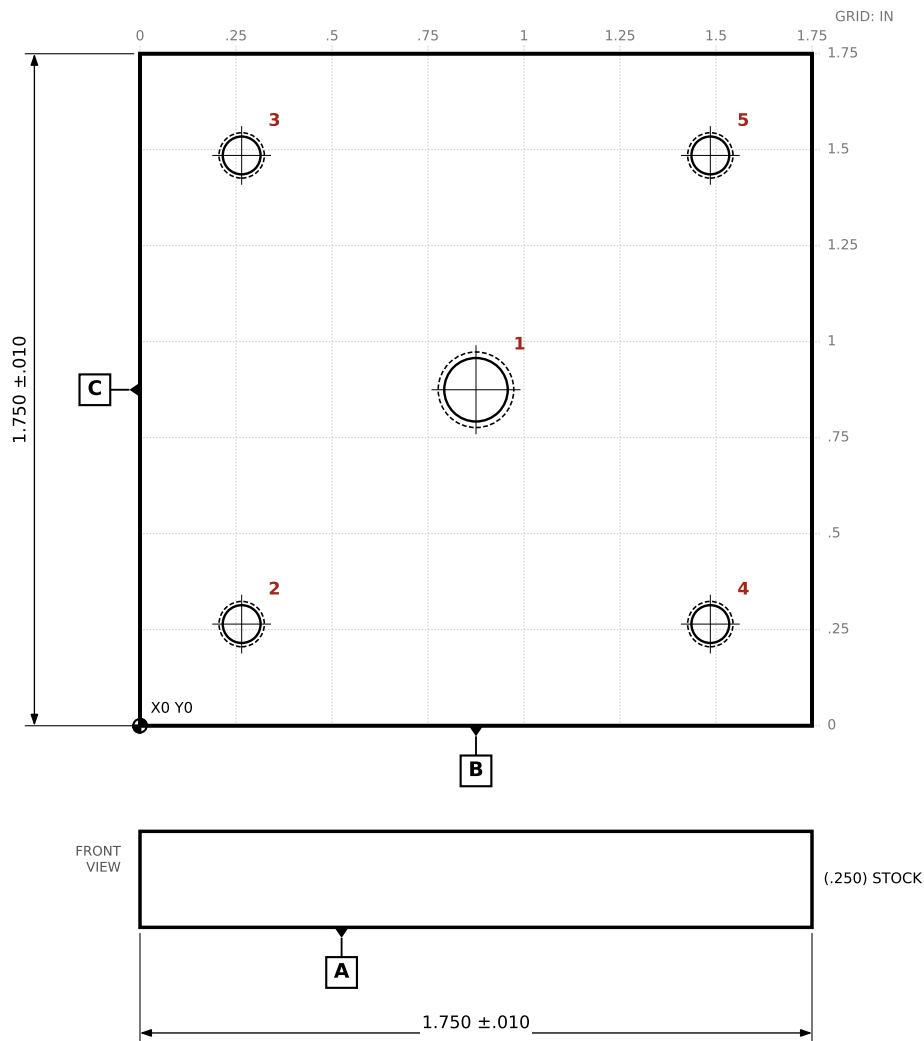
NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X ON THE SHORT EDGES, 1.000 APART, FACE UP.
2. ORIENT: LONG SIDE 1.000 ALONG Y.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = FACE ON THE PARALLELS (OPPOSITE THE MACHINED FACE); B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.
5. SAME BLANK AND HOLES AS THE X CLAMP BAR: RUN THE CLAMP BAR PROGRAM 4 TIMES AND USE 2 AS SPACERS.

PART	X BELT SPACER			
PART NO.	X_BELT_SPACER			
SETUP	OP1			
PROGRAM	O60131_X_BELT_SPACER_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	2	SCALE	2:1	SHEET 18 OF 19
		UNITS	INCH	

PEN PLOTTER: X IDLER CARRIER, OP1

O60141



HOLE TABLE (INCH, PROGRAM COORDINATES)

HOLE	X (BASIC)	Y (BASIC)	METRIC HW	FEATURE	PURPOSE
1	.875	.875		M5x0.8-6H THRU	IDLER AXLE BOLT (M5 SHCS FROM BELOW)
2	.2648	.2648	NEMA17	M3x0.5-6H THRU	IDLER PLATE NEMA17 CLEARANCE HOLE (M3X12 FROM BELOW)
3	.2648	1.4852	NEMA17	M3x0.5-6H THRU	IDLER PLATE NEMA17 CLEARANCE HOLE (M3X12 FROM BELOW)
4	1.4852	.2648	NEMA17	M3x0.5-6H THRU	IDLER PLATE NEMA17 CLEARANCE HOLE (M3X12 FROM BELOW)
5	1.4852	1.4852	NEMA17	M3x0.5-6H THRU	IDLER PLATE NEMA17 CLEARANCE HOLE (M3X12 FROM BELOW)

X FROM DATUM C, Y FROM DATUM B. BOXED VALUES ARE BASIC.

METRIC HW: SET BY A METRIC BOUGHT PART. 4-PLACE VALUES ARE CONVERTED MM.
HOLES 2-5: NEMA 17 SCREWS, 31 MM [1.2205] SQ, CENTERED ON X .875, Y .875.

FEATURE CALLOUTS

HOLE 1

M5x0.8-6H THRU
TAP DRILL Ø.1654

⊕	Ø.020	A	B	C
⊥	Ø.005	Ⓟ 1.160	A	

HOLES 2-5

4X M3x0.5-6H THRU
TAP DRILL Ø.0984

⊕	Ø.005	Ⓟ .250	A	B	C
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UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.XX ±.01 .XXX ±.005 .XXXX ±.0005

ANGLES ±0.5°

BREAK ALL EDGES .005-.015, REMOVE ALL BURRS

MACHINED SURFACES 125 µin Ra MAX

METRIC THREADS PER ASME B1.13M

INTERPRET PER ASME Y14.5-2018

NOTES

1. FIXTURE: SQUARED BLANK ON PARALLELS, VISE JAWS PARALLEL TO X, EITHER PAIR OF EDGES, FACE UP.
2. ORIENT: SQUARE BLANK 1.750 X 1.750.
3. PROGRAM X0 Y0 = LOWER-LEFT CORNER AS DRAWN, Z0 = TOP FACE.
4. DATUMS: A = FACE ON THE PARALLELS, SITS ON THE X IDLER PLATE; B = EDGE AT PROGRAM Y0; C = EDGE AT PROGRAM X0.
5. SITS ON TOP OF THE LEFT X PLATE. MACHINED FACE UP. WITHOUT IT THE X IDLER HAS NO AXLE.

PART	X IDLER CARRIER			
PART NO.	X_IDLER_CARRIER			
SETUP	OP1			
PROGRAM	O60141_X_IDLER_CARRIER_OP1.nc			
MATERIAL	ALUMINUM 6061-T651 PLATE, .250 THK, PER ASTM B209			
FINISH	AS MACHINED			
QTY	1	SCALE	2:1	SHEET 19 OF 19 UNITS INCH